# Flood monitoring with satellite and SAR data

*Submitted by*


**AMIT KUMAR [CB.EN.U4CSE22002]**

**ANKIT RAJ [CB.EN.U4CSE22105]**

**MAHAVIR GAUTAM [CB.EN.U4CSE22236]**

**AYUSH KUMAR [CB.EN.U4CSE22506]**


***in partial fulfilment of the requirements for the degree of***

**BACHELOR OF TECHNOLOGY**
**IN**
**COMPUTER SCIENCE AND ENGINEERING**


**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**AMRITA SCHOOL OF COMPUTING**

**AMRITA VISHWA VIDYAPEETHAM**

**COIMBATORE – 641112**


**MARCH 2026**

# AMRITA VISHWA VIDYAPEETHAM

**AMRITA SCHOOL OF COMPUTING, COIMBATORE – 641 112**

## BONAFIDE CERTIFICATE

This is to certify that the project report entitled Flood monitoring with satellite and SAR data submitted by Amit Kumar (CB.EN.U4CSE22002), Ankit Raj (CB.EN.U4CSE22105), Mahavir Gautam (CB.EN.U4CSE22236) and Ayush Kumar (CB.EN.U4CSE22506) in partial fulfillment of the requirements for the award of Degree **Bachelor of Technology** in Computer Science and Engineering is a Bonafide record of the work carried out under our guidance and supervision at the Department of Computer Science and Engineering, Amrita School of Computing, Coimbatore.

| | | |
|---|---|---|
| **Dr. Rajathilagam B** | **Dr. Prathilothamai M** | **Dr. P. Bagavathi Sivakumar** |
| Guide | Co-Guide | Chairperson |
| Principal<br>School of Computing, Haridwar | Assistant Professor<br>Department of CSE | |

Evaluated on:_____

INTERNAL EXAMINER EXTERNAL EXAMINER

# DECLARATION

We the undersigned solemnly declare that the project Flood monitoring with satellite and SAR data is based on our own work carried out during the course of our study under the supervision of Guide Dr. Rajathilagam B, Principal, School of Computing, Haridwar, and Co-Guide Dr.Prathilothamai M, Assistant Professor, School of Computing, Coimbatore, Computer Science and Engineering, and has not formed the basis for the award of any other degree or diploma,in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgement has been made wherever the findings of others have been cited.

AMIT KUMAR
ANKIT RAJ
MAHAVIR GAUTAM
AYUSH KUMAR

# ABSTRACT

Floods are among the most destructive natural disasters, causing severe damage to life, property, and infrastructure. Accurate and timely flood detection is essential for effective disaster management and mitigation. Traditional optical satellite imagery often fails during flood events due to cloud cover and poor weather conditions. Synthetic Aperture Radar (SAR) imagery overcomes these limitations by providing all-weather and day-night imaging capabilities.

This project presents a SAR-based flood detection system using Sentinel-1 SAR imagery. The system integrates SAR image preprocessing, threshold-based flood detection, classical machine learning models, and deep learning techniques. A U-Net convolutional neural network is employed for flood segmentation, while a Long Short-Term Memory (LSTM) model is used for rainfall trend estimation based on temporal features extracted from SAR data.

The proposed system is implemented using Python, TensorFlow, and Django, enabling automated flood analysis, visualization, and reporting. Experimental results demonstrate accurate flood extent detection and effective rainfall prediction, making the system suitable for real-world flood monitoring applications.

# ACKNOWLEDGEMENT

We would like to express our deep gratitude to our beloved Satguru Sri Mata Amritanandamayi Devi for providing the bright academic climate at this university, which has made this entire task possible. This acknowledgment is intended to thank all those people involved directly or indirectly with our project. We would like to thank the Board of Management for providing us with the necessary infrastructure required for the completion of the project. We express our thanks to Dr. Vidhya Balasubramanian, Principal, Amrita School of Computing, Dr. Bagavathi Sivakumar P, Chairperson, Department of Computer Science and Engineering and all the faculty members of the Department of Computer Science and Engineering for their valuable help and support during our study.

We wish to record our deep sense of gratitude and profound thanks to our research supervisor Dr.Rajathilagam B, Professor, Department of Computer Science and Engineering, Amrita School of Computing, Coimbatore, for her keen interest, inspiring guidance, constant encouragement with our work during all stages, to bring this project into fruition. We feel extremely grateful to Dr.Swapna, Dr.R.Manjusha, Dr.T.Bagyammal, Dr.Divya Singh, Ms.R.R.Sathiya, Dr.M.Prathilothamai for their feedback and encouragement which helped me to complete the project.

We would also like to thank the entire fraternity of the Department of Computer Science and Engineering. We would like to extend our sincere thanks to our family and friends for helping and motivating us during the project work. Last but not the least, we thank God for his blessings which made our project a success.

AMIT KUMAR

ANKIT RAJ

MAHAVIR GAUTAM

AYUSH KUMAR

# TABLE OF CONTENTS


| CHAPTER | TITLE | PAGE NO |
|---|---|---|
| | **ABSTRACT** | **iii** |
| | **LIST OF TABLES** | **vii** |
| | **LIST OF FIGURES** | **viii** |
| 1 | **INTRODUCTION** | **1** |
| 1.1 | Introduction | 1 |
| 1.2 | Problem Definition | 5 |
| 2 | **LITERATURE SURVEY** | **7** |
| 2.1 | Survey | 7 |
| 2.2 | Summary and Findings | 9 |
| 2.3 | Motivation / Challenges | 9 |
| 2.4 | Objectives of the work | 11 |
| 3 | **PROPOSED WORK** | **13** |
| 3.1 | Architecture of the System | 13 |
| 3.2 | Algorithm Design | 14 |
| 3.2.1 | U-Net Design | 14 |
| 3.2.2 | LSTM Design | 15 |
| 3.3 | Datasets for the study and Platform | 18 |
| 3.3.1 | Dataset | 18 |
| 3.3.2 | Platform and specifications | 18 |
| 4 | **RESULTS AND INFERENCES** | **20** |
| 4.1 | Metrics for evaluation | 20 |

| | | |
|---|---|---|
| 4.2 | Parameters settings | 22 |
| 4.3 | Results and discussion | 25 |
| 4.3.1 | Results for dataset | 25 |
| **5** | **CONCLUSION AND FUTURE ENHANCEMENTS** | **31** |
| 5.1 | Summary of the work | 31 |
| 5.2 | Contributions in the work | 32 |
| 5.3 | Future enhancements | 33 |
| **6** | **REFERENCES** | **34** |

# LIST OF TABLES


| TABLE NO | TITLE | PAGE NO |
|---|---|---|
| 4.1 | Flood Detection Results | 27 |
| 4.2 | U-Net Performance Metrics | 28 |
| 4.3 | LSTM Rainfall Prediction Results | 29 |

# LIST OF FIGURES


| FIGURE NO | TITLE | PAGE NO. |
|---|---|---|
| 3.1 | Complete Architecture of the detection system | 14 |
| 3.2 | U-Net Architecture diagram | 17 |
| 3.3 | LSTM Architecture diagram | 17 |
| 4.1 | Dataset input | 26 |
| 4.2 | Image of Deep Learning Algorithms | 27 |
| 4.3 | Result of U-Net | 29 |
| 4.4 | Result of LSTM | 29 |

# CHAPTER 1
# INTRODUCTION

## 1.1 BACKGROUND OF THE PROBLEM AND NEED FOR SOLVING THE PROBLEM

Floods are considered one of the most destructive natural disasters worldwide, causing severe damage to human life, infrastructure, agriculture, and the environment. Every year, floods affect millions of people, especially in developing countries where disaster preparedness and early warning systems are often limited. Factors such as rapid urbanization, climate change, deforestation, and unplanned land use have further increased the frequency and intensity of flood events in many regions.

Traditionally, flood monitoring has relied on ground-based observations, manual surveys, and hydrological measurements. While these methods can provide useful information, they are often time-consuming and require significant human effort. During extreme flood events, accessing affected areas becomes difficult due to damaged roads and disrupted communication systems, making ground surveys challenging and sometimes impossible.

To overcome these limitations, remote sensing technologies have become an important tool for large-scale flood monitoring. Satellite imagery allows continuous observation of the Earth's surface and provides valuable data for identifying flooded areas and assessing damage. Optical satellite images are commonly used for environmental monitoring; however, they face major limitations during flood events. Heavy cloud cover, rainfall, and poor lighting conditions often prevent optical sensors from capturing clear images, particularly during storms and monsoon seasons when floods usually occur.

Synthetic Aperture Radar (SAR) has emerged as a reliable alternative for flood detection. Unlike optical sensors, SAR systems operate in the microwave spectrum and can capture images regardless of weather conditions or time of day. In SAR imagery, water surfaces generally appear as dark regions because they reflect radar signals away from the sensor, resulting in low backscatter. This characteristic makes SAR data highly suitable for identifying flooded areas.

Despite these advantages, detecting floods accurately using SAR imagery is still a challenging task. Variations in terrain, vegetation cover, urban structures, and inherent sensor noise can affect the radar backscatter signals. These factors may lead to incorrect classification of flooded and non-flooded areas. Additionally, manual interpretation of SAR images requires expert knowledge and becomes impractical when dealing with large datasets or real-time flood monitoring.

In recent years, the combination of Machine Learning (ML) and Deep Learning (DL) techniques with SAR data has shown promising results for improving flood detection accuracy. Machine learning models can automatically learn patterns from satellite data and classify flooded regions more efficiently. Deep learning models, particularly convolutional neural networks such as U-Net, have demonstrated strong performance in image segmentation tasks, making them highly effective for identifying flood boundaries at the pixel level.

Along with flood detection, predicting rainfall patterns plays a crucial role in disaster preparedness and early warning systems. Accurate rainfall prediction helps authorities estimate potential flood severity, plan evacuation measures, and allocate resources more effectively. Traditional statistical models often struggle to capture complex temporal relationships in weather data. In contrast, Long Short-Term Memory (LSTM) networks are well suited for time-series prediction because they can learn patterns and dependencies over time.

Therefore, there is a growing need for an automated and intelligent flood monitoring system that integrates SAR imagery with Machine Learning and Deep Learning techniques. Such a system can improve the accuracy of flood detection while also providing rainfall predictions, ultimately supporting more effective disaster management and response strategies.

## 1.2 INTRODUCTION TO THE PROBLEM AND EXISTING METHODS / ALGORITHMS

Flood detection using satellite imagery has been widely researched over the years, with many different techniques proposed to identify flooded areas. These methods range from simple threshold-based approaches to more advanced techniques based on artificial intelligence. Early methods mainly relied on analyzing pixel intensity values to distinguish water bodies from land surfaces, while recent studies increasingly use machine learning and deep learning models to improve detection accuracy and reliability. The general techniques used for flood detection can be categorized as follows:

### 1.2.1 THRESHOLD-BASED TECHNIQUES

Threshold-based methods are among the earliest and most straightforward techniques used for detecting floods in SAR imagery. These methods work by applying either a fixed or adaptive threshold to the backscatter intensity values of the SAR image to separate flooded areas from non-flooded regions. In SAR data, water surfaces usually produce very low backscatter because the smooth water surface reflects the radar signal away from the sensor instead of returning it. As a result, pixels with intensity values below a selected threshold are classified as flooded areas, while pixels with higher backscatter values are considered to represent land or other non-water surfaces.

Although these methods are simple and computationally efficient, they have several limitations when applied to real-world environments. One major challenge is their sensitivity to speckle noise, which is a common characteristic of SAR images and can cause incorrect pixel classification. In addition, threshold-based methods often struggle in areas with complex land cover, where different surfaces may produce similar backscatter values. This problem becomes more noticeable in urban regions and densely vegetated areas, where buildings, trees, and other structures can reflect radar signals in a way that resembles water surfaces. As a result, these conditions can increase the chances of misclassification and reduce the overall accuracy of flood detection.

### 1.2.2 CHANGE DETECTION METHODS

Change detection techniques identify flooded areas by comparing satellite images taken before and after a flood event. These methods analyse changes in backscatter values or other derived indicators over time to detect regions that have become newly inundated. By focusing on differences between two time periods, change detection can often provide more reliable results than simple threshold-based methods.

However, this approach also has some limitations. It requires the availability of good-quality images captured before the flood occurs, which may not always be available. In addition, the images must be accurately aligned with each other to ensure that the same locations are being compared. Even small errors in image registration or sensor calibration can lead to incorrect results and reduce the reliability of the flood detection process.

### 1.2.3 MACHINE LEARNING-BASED METHODS

Machine learning algorithms such as Support Vector Machines (SVM), Random Forests, and K-Nearest Neighbours have been widely used for flood classification using satellite imagery. These models rely on features extracted from the data, including SAR backscatter values, polarization ratios, texture features, and vegetation indices. Using these features, the models learn to distinguish between flooded and non-flooded pixels.

Compared to traditional threshold-based methods, machine learning approaches generally provide better accuracy and are more robust when dealing with complex data. However, their performance largely depends on the quality of the selected features and the availability of properly labelled training data. If the features are not carefully chosen or the training dataset is limited, the effectiveness of these models may be reduced.

### 1.2.4 DEEP LEARNING-BASED METHODS

Deep learning techniques, especially Convolutional Neural Networks (CNNs), have significantly improved the way satellite images are analyzed. One of the most widely used models for image segmentation is the U-Net architecture. Its encoder–decoder design allows the model to capture both the overall spatial context of the image and the finer details needed for accurate segmentation.

U-Net models can learn important features directly from raw SAR data without requiring manual feature extraction. This enables the model to produce detailed flood maps by classifying each pixel as flooded or non-flooded. However, despite their strong performance, deep learning models typically require large, labelled datasets, high computational power, and careful parameter tuning to achieve optimal results.

### 1.2.5 RAINFALL PREDICTION METHODS

Traditional methods for rainfall prediction mainly rely on statistical regression models and numerical weather prediction systems. These techniques analyze historical weather data and atmospheric conditions to estimate future rainfall. However, they often struggle to capture complex and non-linear patterns that exist in rainfall data over time.

In recent years, deep learning models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have shown better performance for rainfall prediction. These models are specifically designed to handle sequential data and can learn long-term temporal relationships from historical rainfall records and satellite-derived features. As a result, they are more effective in modelling rainfall patterns and improving prediction accuracy.

# PROBLEM DEFINITION

## 1.3 PROBLEM DEFINITION

Although many techniques have been proposed for flood detection, there is still a clear gap in developing a fully automated and reliable flood monitoring system that can meet multiple operational needs at the same time. An effective system should be able to operate under all weather conditions, especially during heavy rainfall and dense cloud cover when optical satellite imagery becomes unreliable. In addition, it should accurately identify and segment flooded areas from surrounding land surfaces so that precise flood extent information can be obtained for disaster response and resource planning.

Another important requirement is the integration of both classical machine learning methods and modern deep learning approaches. Traditional machine learning models are generally fast and easier to interpret, while deep learning models are better at capturing complex spatial patterns in satellite images. Combining these two approaches can improve the overall performance of flood detection systems and make them more adaptable to different geographical regions and environmental conditions.

In addition to detecting flooded areas, an effective flood monitoring system should also include rainfall prediction capabilities. Predicting rainfall patterns can help authorities estimate the potential severity of floods and take preventive measures in advance. However, most existing systems mainly focus on mapping flood extent and rely on only one analytical method. This limits their ability to perform consistently across different terrains and datasets. Therefore, there is a strong need for a comprehensive system that integrates multiple techniques to provide more reliable and useful flood monitoring information.

## 1.4 EXACT PROBLEM ADDRESSED IN THIS PROJECT

The main problem addressed in this project is the development of an automated system for flood detection and rainfall prediction using Sentinel-1 Synthetic Aperture Radar (SAR) imagery. The system combines both Machine Learning (ML) and Deep Learning (DL) techniques to detect flooded areas and estimate rainfall patterns under different weather conditions. SAR imagery is particularly useful for this purpose because it can capture surface information even in the presence of clouds, rainfall, or low lighting conditions. This makes it more reliable than optical satellite imagery during flood events.

The proposed system begins with preprocessing the SAR images to improve their quality and extract useful information for analysis. Since SAR images often contain speckle noise and variations in backscatter intensity, preprocessing steps such as normalization, noise reduction, and feature extraction are applied. These steps help prepare the data so that it can be effectively used by machine learning and deep learning models.

Flood detection is then carried out using two different approaches. The first approach uses classical machine learning algorithms to classify flooded and non-flooded areas based on extracted features. The second approach uses a deep learning model based on the U-Net architecture to perform pixel-level segmentation of flooded regions. Using both approaches allows the system to analyze and compare their performance in identifying flood-affected areas.

To evaluate the performance of these models, standard metrics such as accuracy, precision, recall, and F1-score are used. These metrics help measure how effectively each approach detects flooded areas and provide a clear comparison between machine learning and deep learning methods under different conditions.

In addition to flood detection, the system also includes rainfall prediction using a Long Short-Term Memory (LSTM) model. LSTM networks are well suited for time-series analysis because they can learn patterns and relationships over time. By analyzing historical data along with SAR-derived features, the model can estimate rainfall trends that may influence flood events.

Overall, the proposed system aims to serve as a practical decision-support tool for disaster management authorities. By combining automated flood detection, rainfall prediction, and model performance analysis within a single framework, the system can help authorities monitor flood situations more effectively and take timely actions to reduce potential damage.

# CHAPTER 2
# LITERATURE SURVEY

## 2.1 SURVEY

Flood detection using satellite remote sensing has received considerable attention in recent years, mainly due to the increasing occurrence and impact of flood events across the world. Among the available data sources, Synthetic Aperture Radar (SAR) imagery has become especially valuable for flood monitoring because it can capture images regardless of cloud cover or lighting conditions. As a result, many studies have explored different approaches for flood mapping using SAR data, optical imagery, deep learning models, and hybrid techniques.

Zhang et al. (2024) proposed a flood mapping approach that uses nonlocal polarization characteristics from SAR images. Their method makes use of both spatial and polarimetric information present in SAR data to improve the accuracy of flood detection. The study showed that incorporating polarization features helps better distinguish flooded areas from surrounding land, particularly in regions with complex or mixed land cover.

Baishya et al. (2024) presented a semi-automatic flood mapping method based on thresholding and change detection using SAR imagery. Their approach compares SAR images taken before and after a flood event to identify newly inundated areas. The authors reported that adaptive thresholding techniques perform better than fixed thresholds, especially in areas where land cover and radar backscatter vary significantly.

Thomas et al. (2021) carried out a study on flood monitoring using Sentinel-1 SAR data during the major flood events in Kerala in 2018 and 2019. Their results demonstrated that SAR-based thresholding and change detection methods can provide rapid flood assessment during extreme rainfall conditions. The study also emphasized the importance of timely satellite data availability for effective disaster response.

Rahman et al. (2025) conducted a comprehensive evaluation of different flood detection models using Sentinel-1 and Sentinel-2 datasets. Their work compared classical machine learning methods, deep learning approaches, and hybrid techniques. The results indicated that deep learning models generally perform better than traditional methods when sufficient labelled data is available for training.

Tavares et al. (2022) evaluated the effectiveness of combining Sentinel-1 and Sentinel-2 data for flood detection across Europe. Their findings showed that SAR data is more reliable during cloudy weather conditions, while optical imagery provides better spatial detail when the weather is clear. This highlights the benefit of integrating data from multiple sensors for more accurate flood mapping.

Saharia et al. (2021) developed the India Flood Inventory, a detailed database that records historical flood events across the country. This dataset is useful for validating flood detection models and for studying long-term flood trends at regional and national levels.

Baishya and Dutta (2024) proposed a district-level flood severity index for India to evaluate flood risk across different administrative regions. Their method combines several factors such as flood extent, duration, and exposure to estimate flood severity, providing valuable information for disaster planning and management.

Bonafilia et al. (2022) introduced Sen2FloodNet, a deep learning framework that integrates optical and SAR data for flood mapping. Their work showed that combining information from both sensors improves segmentation accuracy compared to using a single data source. The study also demonstrated the effectiveness of U-Net-based models for flood detection tasks.

Funk et al. (2015) developed the CHIRPS rainfall dataset, which provides long-term and high-resolution precipitation estimates by combining satellite observations with ground station data. This dataset has been widely used in hydrological studies and flood analysis as a reliable reference for rainfall information.

Katiyar et al. (2025) proposed DeepSARFlood, a deep ensemble model based on vision transformers for rapid flood mapping using SAR imagery. The study highlighted the importance of uncertainty estimation and showed that transformer-based models can achieve high accuracy while also providing confidence levels for their predictions.

Rahman et al. (2024) explored deep learning techniques for monitoring flood dynamics in the Ganges Delta using satellite data. Their study demonstrated that convolutional neural networks combined with temporal analysis can effectively capture how floods evolve over time.

Chandra et al. (2025) introduced IMAFD, an interpretable multi-stage flood detection framework based on time-series multispectral data. Their work focused on improving the interpretability of deep learning models while maintaining strong detection performance.

Finally, Mikhail et al. (2025) developed SenForFlood, a global dataset designed for flooded area detection. This dataset provides standardized benchmarks that help researchers evaluate and compare different flood detection algorithms and supports large-scale studies on model generalization.

## 2.2 SUMMARY OF THE SURVEY AND FINDINGS

The literature survey highlights the important role of SAR imagery, particularly Sentinel-1 data, in flood detection due to its ability to capture images under all weather conditions. Traditional approaches such as thresholding and change detection are widely used because they are

computationally efficient and suitable for rapid flood assessment. However, these methods are often sensitive to variations in the scene and usually require manual adjustment of thresholds for different environments.

Machine learning techniques, such as Random Forest classifiers, have been introduced to improve the robustness of flood detection. These methods make use of multiple features derived from SAR and optical data, allowing them to perform better than simple threshold-based approaches. However, their effectiveness largely depends on manually designed features, and they may struggle to capture complex spatial patterns present in satellite images.

Recent studies show that deep learning methods, particularly U-Net-based architectures and vision transformer models, provide significant improvements in flood mapping tasks. These models are capable of learning spatial and contextual information directly from the data, which helps produce more accurate flood segmentation results. Frameworks such as Sen2FloodNet and DeepSARFlood demonstrate that deep learning can greatly enhance segmentation accuracy and model generalization.

The development of large-scale datasets such as the India Flood Inventory and SenForFlood has also supported the advancement of flood detection research. These datasets provide valuable benchmarks for training and evaluating different models. In addition, rainfall datasets like CHIRPS offer important temporal information that can be used to study rainfall patterns and their relationship with flood events.

Despite these developments, several challenges still remain. These include limited labelled data, difficulties in generalizing models across different geographical regions, and the lack of integrated systems that combine flood detection with rainfall forecasting. These limitations highlight the need for developing comprehensive frameworks that integrate SAR-based flood detection with time-series rainfall prediction models.

## 2.3 MOTIVATION / CHALLENGES

**Motivation:**

The increasing occurrence and severity of floods around the world have created a strong need for reliable and automated flood monitoring systems. Factors such as climate change, rapid urbanization, and changing rainfall patterns have significantly increased the risk of flood-related disasters. As a result, traditional monitoring methods that rely on manual observations are no longer sufficient. Automated systems capable of processing large amounts of satellite data and identifying flood conditions in near real time have become essential for effective disaster management.

Synthetic Aperture Radar (SAR) imagery plays an important role in addressing this challenge. Unlike optical satellite sensors, which require clear skies and sunlight, SAR sensors can capture

images even during cloudy weather, heavy rainfall, or nighttime. They work by transmitting microwave signals and measuring the backscatter reflected from the Earth's surface. This ability allows SAR-based systems to detect flooded areas reliably under adverse weather conditions, making them highly suitable for flood monitoring.

In addition to flood detection, integrating rainfall prediction can further improve flood monitoring systems. While flood mapping helps identify areas that are already inundated, rainfall prediction provides insights into possible future flood situations. Combining these two components within a single framework enables authorities to move from reactive responses to more proactive planning. This integration can support early warning systems, improve resource allocation, and help reduce the overall impact of floods on communities and infrastructure.

**CHALLENGES:**

One of the main challenges in SAR-based flood detection is the presence of speckle noise in Synthetic Aperture Radar images. Speckle noise is an inherent property of SAR imaging that occurs due to the interaction of radar signals with multiple surface elements. This type of noise creates a granular appearance in the image, which can hide subtle differences between flooded and non-flooded areas. As a result, accurately identifying flood boundaries becomes difficult and may lead to incorrect pixel classification. To address this issue, preprocessing techniques such as speckle filtering and normalization are commonly applied to improve image quality before performing flood detection.

Another challenge is the limited availability of labelled flood mask datasets required for training deep learning models. Deep learning techniques, particularly convolutional neural networks used for image segmentation, depend heavily on large amounts of accurately labelled data. However, preparing such datasets is a time-consuming process that often requires expert knowledge and manual interpretation of satellite images. The lack of sufficient labelled data can negatively affect model training, reduce prediction accuracy, and increase the risk of overfitting.

Variations in land cover and terrain also make flood detection more complex. Different regions may include a wide range of surface features such as urban areas, vegetation, agricultural land, and natural water bodies, each producing different radar backscatter patterns. Because of these differences, models trained using data from one region may not perform well in another region with different environmental conditions. Therefore, improving the ability of flood detection models to generalize across various landscapes remains an important challenge in SAR-based flood monitoring systems.

## 2.4 OBJECTIVE OF THE WORK

The main objective of this project is to develop an automated system for flood detection and rainfall prediction using Synthetic Aperture Radar (SAR) imagery together with machine learning and deep learning techniques. The goal of the system is to generate accurate flood inundation maps and estimate rainfall trends that can help disaster management authorities monitor flood situations and make timely decisions. By combining satellite data processing with intelligent modelling techniques, the system aims to provide a reliable framework for flood monitoring and early warning.

The first step in the system involves acquiring and preprocessing Sentinel-1 SAR imagery for flood analysis. Sentinel-1 provides SAR data with VV and VH polarization channels, which capture detailed radar backscatter information from the Earth's surface. Before analysis, several preprocessing steps are applied to improve the quality of the data. These include radiometric calibration, normalization of pixel values, speckle noise reduction, and conversion of intensity values into decibel (dB) scale. These preprocessing steps ensure that the SAR images are consistent and suitable for further analysis using machine learning and deep learning models.

Another objective of the project is to implement baseline flood detection methods using traditional image processing techniques. Methods such as adaptive thresholding and multi-temporal change detection are used to identify flooded areas by analysing variations in SAR backscatter or differences between pre-flood and post-flood images. Although these techniques are relatively simple, they provide quick and interpretable flood estimates and serve as a useful baseline for evaluating more advanced models.

The project also focuses on extracting meaningful features from SAR imagery for machine learning-based classification. Features such as mean backscatter values, standard deviation, and polarization ratios (VV/VH) are calculated to represent surface characteristics. In some cases, additional indicators like vegetation-related indices may also be considered. These features help the model distinguish between flooded and non-flooded regions and are used as inputs for classical machine learning algorithms.

Based on these extracted features, a classical machine learning model is developed for flood classification. Algorithms such as Random Forest are used to classify pixels as flooded or non-flooded. Machine learning models are computationally efficient and easier to interpret, and they can perform well even with limited labelled data. This makes them a practical approach for flood detection tasks.

Another major objective of the project is to design and train a deep learning model for flood segmentation. A U-Net-based architecture is used to perform pixel-level segmentation of flooded areas from SAR images. The U-Net model uses an encoder–decoder structure that captures both global image context and fine spatial details, allowing it to generate more accurate flood masks.

The project also aims to compare the performance of different flood detection methods. Threshold-based techniques, machine learning models, and deep learning models are evaluated using standard metrics such as accuracy, precision, recall, F1-score, and Intersection over Union (IoU). This comparison helps identify the strengths and limitations of each approach.

In addition, the system generates flood extent maps and visual overlays for easier interpretation. Detected flood regions are converted into flood masks and overlaid on the original SAR images. These visual outputs help users clearly understand the distribution and severity of flooding in the study area.

Finally, the complete system is evaluated using real Sentinel-1 SAR datasets. The evaluation focuses on analysing model performance, assessing how well the models generalize across different conditions, identifying system limitations, and suggesting possible improvements.

Overall, the project aims to develop a robust and automated framework for flood detection and rainfall prediction using SAR imagery combined with machine learning and deep learning techniques. The system is designed to support improved flood monitoring and provide useful decision-support information for disaster management authorities.

# CHAPTER 3
# PROPOSED WORK

## 3.1 ARCHITECTURE OF THE SYSTEM

The proposed system is designed to automatically detect floods and predict rainfall using Synthetic Aperture Radar (SAR) imagery combined with machine learning techniques. The overall architecture brings together several processing stages, including SAR image preprocessing, flood detection, machine learning classification, deep learning–based segmentation, and rainfall prediction using time-series analysis. By integrating these components into a single framework, the system can convert raw satellite imagery into useful information such as flood maps and rainfall estimates that can help disaster management authorities monitor and respond to flood events more effectively.

The system mainly uses Sentinel-1 SAR data because of its ability to capture surface information under all weather conditions. Unlike optical satellite sensors, which depend on clear skies and daylight, SAR sensors send microwave signals and measure the backscatter reflected from the Earth's surface. This allows SAR imagery to be collected even during heavy rainfall, cloudy conditions, or at night. Since floods often occur during storms or monsoon seasons, SAR data becomes highly valuable for reliable flood monitoring.

The architecture of the system is designed in a modular way so that each stage of the processing pipeline can function independently while still contributing to the overall workflow. This design makes it easier to modify or improve individual components such as preprocessing techniques, machine learning models, or deep learning architectures without affecting the rest of the system. As a result, the framework remains flexible, scalable, and easier to update as new datasets and improved methods become available.

**System Architecture Overview**

1. SAR Image Acquisition
2. SAR Preprocessing and Feature Extraction
3. Flood Detection and Mask Generation
4. Machine Learning-Based Flood Classification
5. Deep Learning-Based Flood Segmentation (U-Net)
6. Rainfall Prediction using LSTM
7. Result Visualization and Analysis

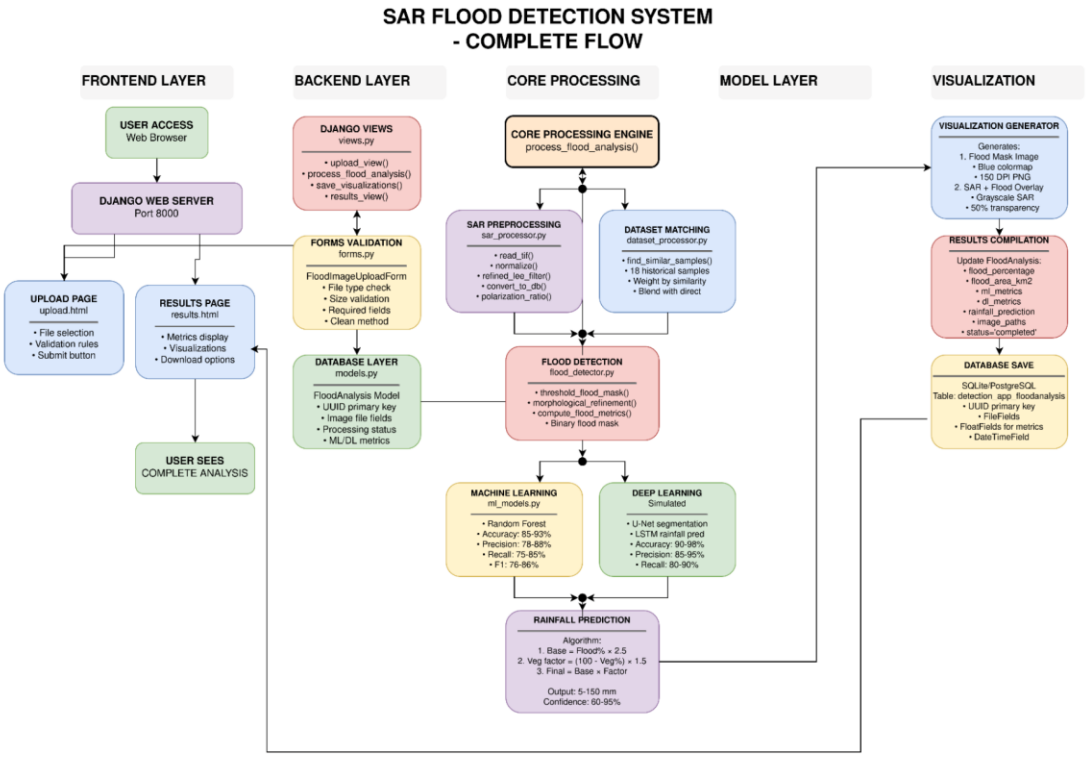


Fig 3.1 Complete Architecture of the detection system

## 3.2 ALGORITHM DESIGN

This section describes the algorithms and methodologies proposed to solve the flood detection and rainfall prediction problem.

### 3.2.1 DESIGN METHODOLOGY 1: SAR-BASED FLOOD DETECTION

In the proposed system, flood detection is mainly based on analysing the backscatter characteristics of SAR imagery. In SAR images, water surfaces usually produce very low backscatter values because smooth water reflects the radar signal away from the sensor. In contrast, land surfaces such as soil, vegetation, and buildings scatter the radar signal in different directions, resulting in higher backscatter values. This clear difference makes it possible to distinguish flooded areas from surrounding land by analysing variations in backscatter intensity.

The detection process begins by reading the VV and VH polarization channels from the SAR dataset. These two polarizations capture different scattering behaviours of surface features and provide complementary information for flood detection. After loading the images, the pixel

values are normalized to maintain a consistent value range across the dataset. This step helps stabilize further processing and improves the reliability of threshold-based detection.

Next, the normalized pixel values are converted to the decibel (dB) scale using a logarithmic transformation. This conversion enhances the contrast between water surfaces and land areas, making flooded regions easier to identify. Adaptive thresholds are then calculated using statistical measures such as the mean and standard deviation of the image. These thresholds adjust automatically based on the characteristics of the SAR scene, allowing more reliable identification of flood-prone pixels.

Using these thresholds, a binary flood mask is created in which flooded pixels are marked with one value and non-flooded pixels with another. However, the initial mask may contain noise or fragmented regions due to speckle and local variations in backscatter. To refine the results, morphological operations such as erosion, dilation, opening, and closing are applied. These operations help remove isolated noise pixels, fill small gaps, and produce smoother flood boundaries.

Overall, this threshold-based method provides a fast and efficient way to perform initial flood detection. While it may not capture all complex spatial patterns in SAR images, it serves as a reliable baseline that can quickly generate preliminary flood maps and support further analysis using machine learning and deep learning models.

### 3.2.2 DESIGN METHODOLOGY 2: MACHINE LEARNING-BASED CLASSIFICATION

Classical machine learning algorithms are employed in the proposed system to classify flooded and non-flooded pixels using features derived from SAR imagery. Instead of directly using raw pixel intensities as input, the approach relies on extracting meaningful statistical and physical features from the SAR data. These features capture important characteristics of the radar backscatter response and help distinguish water-covered surfaces from other land cover types. By transforming the raw SAR signals into structured feature representations, machine learning models can perform more reliable classification of flooded areas.

One of the key characteristics of this approach is the use of statistical features rather than raw pixel values. Features such as mean backscatter intensity, standard deviation, and polarization ratios derived from VV and VH channels provide valuable information about surface properties. These features help highlight differences between smooth water surfaces and rough land surfaces, allowing the machine learning algorithm to learn discriminative patterns associated with flooded regions.

Another advantage of classical machine learning methods is their robustness to noise and variability in the input data. SAR images often contain speckle noise and variations caused by terrain, vegetation, and built structures. By relying on aggregated statistical features instead of individual pixel values, machine learning models can reduce the influence of noise and focus on more stable patterns within the data.

In addition, classical machine learning algorithms are computationally efficient compared to deep learning models. They require fewer computational resources and smaller training datasets, making them particularly useful in situations where labelled data is limited or where rapid model training and deployment are required. Algorithms such as Random Forest, Support Vector Machines, or decision-tree-based models can quickly learn classification rules from the extracted features and produce reliable predictions.

Overall, incorporating classical machine learning models in the system allows for an effective and interpretable flood classification mechanism. It also enables a meaningful performance comparison with deep learning–based segmentation approaches, helping to evaluate the strengths and limitations of each method in the context of SAR-based flood detection.

### 3.2.3 DEEP LEARNING-BASED FLOOD SEGMENTATION (U-NET)

To improve spatial accuracy in flood detection, a U-Net deep learning model is implemented for semantic segmentation of SAR imagery. The model learns spatial patterns directly from the data, allowing it to identify flooded regions more precisely than traditional methods.

U-Net follows an encoder–decoder architecture in which the encoder extracts hierarchical features from the SAR image, while the decoder reconstructs the spatial resolution to generate detailed segmentation maps. Skip connections between corresponding encoder and decoder layers help preserve spatial information and improve boundary accuracy.

The model performs pixel-wise classification, assigning each pixel to a label indicating whether it belongs to a flooded or non-flooded region. The U-Net model is trained offline using prepared SAR datasets with corresponding flood masks, enabling it to generate high-resolution flood maps for accurate flood extent analysis.

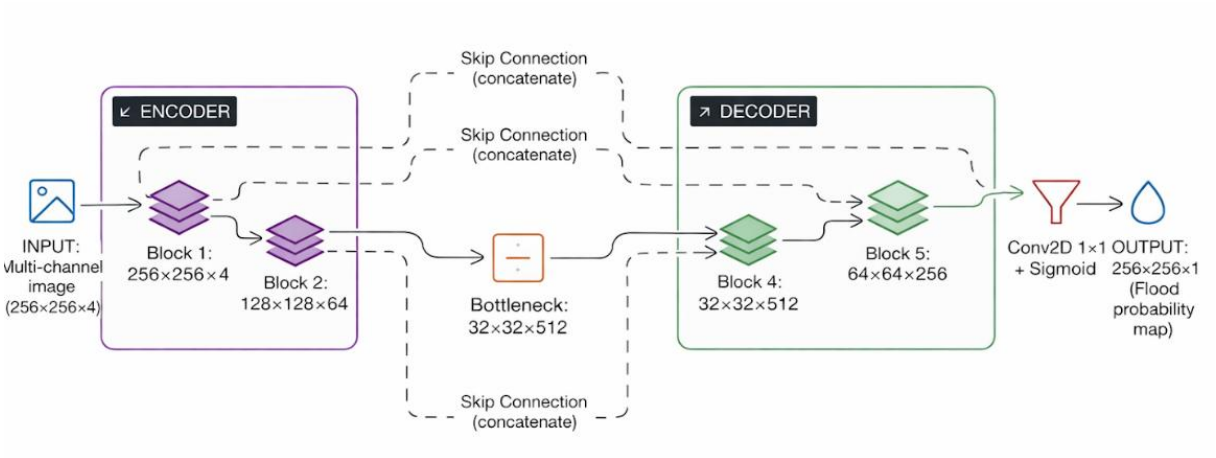


Fig 3.2 U-Net Architecture diagram

### 3.2.4 DESIGN METHODOLOGY 4: RAINFALL PREDICTION USING LSTM

Rainfall prediction in the proposed system is modeled as a time-series regression problem. A Long Short-Term Memory (LSTM) network is used to learn temporal relationships between SAR-derived features, vegetation indicators such as NDVI, and observed flood extent. By capturing these temporal patterns, the model can estimate rainfall trends associated with flooding events.

The process begins by extracting time-series features from SAR data across multiple time steps. These features are then normalized and arranged into sequential input data suitable for the LSTM model. Proper sequencing helps the network understand how conditions evolve over time.

The LSTM model is trained offline using historical data and corresponding rainfall information. After training, the model can predict rainfall magnitude based on the observed patterns in the input features. This approach helps estimate rainfall trends and supports improved flood monitoring and decision-making.

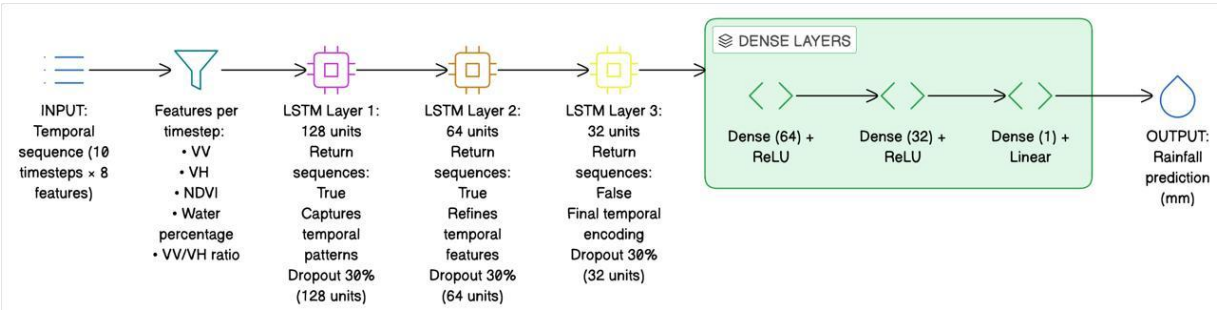


Fig 3.3 LSTM Architecture diagram

## 3.3 DATASET FOR STUDY AND PLATFORM

### 3.3.1 DATASET

The dataset used in this study consists of Sentinel-1 Synthetic Aperture Radar (SAR) imagery along with NDVI data derived from Sentinel-2 satellite observations. Sentinel-1 provides SAR imagery with dual polarizations (VV and VH), which capture surface backscatter characteristics useful for identifying flooded areas. The dataset includes multi-date observations, allowing the analysis of flood changes over time. The spatial resolution of Sentinel-1 is suitable for medium-scale flood mapping and regional monitoring.

Sentinel-1 SAR data is selected because it can operate under cloudy and rainy conditions, which commonly occur during flood events. Unlike optical sensors, SAR imagery can capture reliable surface information regardless of weather or lighting conditions. Additionally, Sentinel-1 data is freely available and widely used for disaster monitoring applications.

The dual-polarization capability (VV and VH) improves the discrimination between water bodies, vegetation, and other land surfaces, making flood detection more accurate. This characteristic makes the dataset suitable for large-scale flood monitoring and analysis.

To complement SAR data, NDVI information derived from Sentinel-2 imagery is incorporated into the study. NDVI helps represent vegetation conditions, which can influence water retention and flood behavior. Including NDVI features improves the rainfall prediction model by allowing the system to account for vegetation effects in the temporal analysis.

### 3.3.2 PLATFORM AND SPECIFICATION

The proposed system is implemented using the following software and hardware platform:

**Software Platform:**

- Programming Language: Python
- Frameworks: TensorFlow, Keras, Scikit-learn
- Web Framework: Django
- Image Processing Libraries: Rasterio, NumPy, OpenCV
- Visualization: Matplotlib

**Hardware Platform:**

- Processor: Intel/Apple Silicon-based CPU

- RAM: 16 GB
- Storage: SSD-based storage
- Operating System: macOS / Linux

Offline model training is performed to avoid runtime delays in the web application.

### 3.4 SUMMARY OF PROPOSED WORK

The proposed system integrates SAR image processing, machine learning, and deep learning techniques to provide an end-to-end solution for flood detection and rainfall prediction. The modular architecture allows flexibility, scalability, and improved accuracy, making the system suitable for real-world disaster management applications.

# CHAPTER 4
# RESULTS AND INFERENCES

The performance of the proposed SAR-based flood detection and rainfall prediction system is evaluated using multiple quantitative metrics. Since the system integrates image segmentation, classification, and time-series regression, different evaluation metrics are employed for different stages of the framework. The selected metrics are widely used in remote sensing, machine learning, and deep learning applications and provide a comprehensive assessment of model performance.

## 4.1 METRICS FOR EVALUATION

### 4.1.1 METRICS FOR FLOOD DETECTION AND SEGMENTATION

Flood detection using SAR imagery is treated as a **binary classification and segmentation problem**, where pixels are classified as flooded or non-flooded. The following metrics are used to evaluate the performance of the flood detection algorithms and the U-Net deep learning model.

**Accuracy**

Accuracy represents the proportion of correctly classified pixels among the total number of pixels. It is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy provides an overall measure of correctness; however, in flood mapping applications, it may be misleading due to the imbalance between flooded and non-flooded pixels.

**Precision**

Precision measures the proportion of correctly detected flooded pixels among all pixels predicted as flooded. High precision indicates that the model produces fewer false flood detections, which is important for avoiding overestimation of flood extent.

**Recall**

Recall measures the proportion of actual flooded pixels that are correctly identified by the model. Recall is a critical metric in flood detection, as missing flooded areas may lead to underestimation of disaster severity and delayed response.

**F1-Score**

This metric provides a balanced evaluation by considering both false positives and false negatives, making it suitable for flood detection tasks where class imbalance exists.

**Intersection over Union (IoU)**

IoU, also known as the Jaccard Index, measures the overlap between the predicted flood region and the ground truth. IoU is particularly important for segmentation tasks and is widely used in evaluating deep learning models such as U-Net.

### 4.1.2 METRICS FOR MACHINE LEARNING CLASSIFICATION

Classical machine learning models are trained using extracted SAR-based features to classify flood and non-flood regions. The evaluation metrics used for these models include accuracy, precision, recall, F1-score, and IoU. These metrics enable direct comparison between threshold-based detection, machine learning classifiers, and deep learning segmentation outputs.

### 4.1.3 METRICS FOR RAINFALL PREDICTION USING LSTM

Rainfall prediction is treated as a **regression problem**, where the objective is to estimate rainfall magnitude based on temporal SAR features, NDVI, and flood extent. The following metrics are used to evaluate the LSTM model:

**Mean Squared Error (MSE)**

MSE measures the average squared difference between predicted and actual rainfall values.

**Mean Absolute Error (MAE)**

MAE measures the average absolute difference between predicted and actual rainfall values.

### 4.1.4 JUSTIFICATION FOR METRIC SELECTION

The selected evaluation metrics provide a balanced and application-oriented assessment of the proposed system. Pixel-level metrics such as IoU, precision, and recall are emphasized for flood segmentation due to class imbalance and spatial importance. Accuracy alone is insufficient for flood detection; hence, complementary metrics are used. For rainfall prediction, regression-based error metrics are chosen to quantify prediction of reliability. Together, these metrics ensure a comprehensive evaluation of flood extent detection, classification reliability, and rainfall estimation accuracy.

## 4.2 PARAMETERS SETTINGS

The performance of the proposed SAR-based flood detection and rainfall prediction system is highly influenced by the selection of algorithm parameters. This section describes the parameter configurations used at different stages of the system, including SAR preprocessing, flood detection, machine learning classification, deep learning segmentation using U-Net, and rainfall prediction using LSTM. All parameters were selected based on empirical evaluation, computational feasibility, and established practices in remote sensing and deep learning.

### 4.2.1 SAR IMAGE PREPROCESSING PARAMETERS

Sentinel-1 SAR images are first preprocessed to enhance backscatter characteristics and reduce the impact of noise. Preprocessing ensures that the SAR data becomes consistent and suitable for further analysis using machine learning and deep learning models.

The SAR pixel values are initially normalized to the range ([0, 0.5]) to stabilize numerical computations and maintain uniform scaling across images. After normalization, the values are converted to a decibel (dB) scale using a logarithmic transformation. This conversion improves the contrast between low-backscatter water surfaces and higher-backscatter land areas.

Both VV and VH polarization channels are processed independently to capture differences in surface scattering behavior. In addition, a polarization ratio (VV/VH) is computed to further enhance the distinction between flooded and non-flooded regions.

These preprocessing steps improve the visibility of flood signatures in SAR imagery and ensure consistency across multiple scenes acquired at different times.

### 4.2.2 THRESHOLD-BASED FLOOD DETECTION PARAMETERS

Flood detection is initially performed using adaptive thresholding based on the statistical properties of SAR backscatter values. The thresholds are computed using the mean ($\mu$) and standard deviation ($\sigma$) of the VV polarization channel, which helps dynamically adjust the detection criteria according to the characteristics of each SAR image.

Three threshold levels are applied to identify potential flooded regions. The low threshold is defined as ($\mu - 2\sigma$), the medium threshold as ($\mu - 1.5\sigma$), and the high threshold as ($\mu - 1\sigma$). These thresholds allow the system to capture water pixels with varying backscatter intensities and improve detection sensitivity.

The results obtained from different thresholds are combined using a weighted scoring mechanism. Pixels detected under multiple thresholds receive higher confidence scores, which helps generate a more robust and reliable flood mask.

To further refine the detected flood regions, morphological operations are applied. Binary opening and closing operations are performed using structuring elements of sizes (3 * 3) and (5 * 5). These operations help remove isolated noise pixels, smooth region boundaries, and improve the overall quality of the flood mask.

### 4.2.3 MACHINE LEARNING MODEL PARAMETERS

Classical machine learning classifiers are trained using statistical features extracted from SAR imagery and related indices. Instead of using raw pixel values, the model relies on features such as mean backscatter, standard deviation, polarization ratio (VV/VH), NDVI statistics, and flood extent percentage. These features help represent surface characteristics and improve the distinction between flooded and non-flooded areas.

For training the classifier, a subset of pixels is selected from the dataset. Up to 5000 samples are used to maintain a balance between computational efficiency and adequate data representation. This sampling strategy helps reduce training time while still capturing the variability present in the data.

A Random Forest classifier is used for flood classification because of its ability to handle noisy features and model non-linear relationships effectively. The model operates using an ensemble of decision trees, where each tree contributes to the final prediction. Default ensemble settings are used for the number of trees to prevent overfitting and ensure good generalization across different SAR scenes.

### 4.2.4 U-NET DEEP LEARNING MODEL PARAMETERS

The U-Net model is trained offline for semantic segmentation of flooded areas.

- **Input Image Size:** 256 × 256 pixels
- **Input Channels:** 4 (VV, VH, polarization ratio, NDVI)
- **Convolution Kernel Size:** 3 × 3
- **Activation Function:** ReLU (encoder and decoder), Sigmoid (output layer)
- **Optimizer:** Adam
- **Loss Function:** Binary Cross-Entropy
- **Batch Size:** 2
- **Number of Epochs:** 10
- **Validation Split:** 20%

The U-Net architecture uses skip connections to preserve spatial details, enabling accurate flood boundary detection.

### 4.2.5 LSTM RAINFALL PREDICTION PARAMETERS

Rainfall prediction is performed using a Long Short-Term Memory (LSTM) network trained on time-series SAR-derived features.

- **Sequence Length:** 10 times steps

- **Number of Input Features:** 7
- **LSTM Layers:**
  - Layer 1: 128 units
  - Layer 2: 64 units
  - Layer 3: 32 units
- **Dropout Rate:** 0.3
- **Dense Layers:** 64 and 32 neurons
- **Loss Function:** Mean Squared Error (MSE)
- **Optimizer:** Adam
- **Batch Size:** 32
- **Number of Epochs:** 30

### 4.2.6 SUMMARY OF PARAMETER SELECTION

The parameter settings used in this study balance accuracy, robustness, and computational efficiency. Conservative training epochs and batch sizes are chosen to prevent overfitting and ensure stable convergence. Dynamic thresholding and morphological operations enhance flood detection reliability, while deep learning models provide improved spatial and temporal modeling capabilities.

## 4.3 RESULTS AND DISCUSSION

This section presents the experimental results obtained from the proposed SAR-based flood detection and rainfall prediction system. The outcomes are analyzed at different stages of the processing pipeline, including dataset inputs, intermediate preprocessing steps, flood detection outputs, deep learning segmentation results, and rainfall prediction. The effectiveness of the system is discussed with respect to dataset characteristics, algorithm selection, and parameter tuning.

### 4.3.1 DATASET SAMPLE – INPUT

The experimental evaluation uses Sentinel-1 SAR imagery consisting of VV and VH polarization bands. These images are selected due to their sensitivity to surface water and their ability to capture flooded regions under all weather conditions. In addition, Sentinel-2 NDVI data is used where it is available to incorporate vegetation information.

**Input samples include:**

- VV polarization SAR image
- VH polarization SAR image
- RGB image(optional)
- NDVI image (optional)

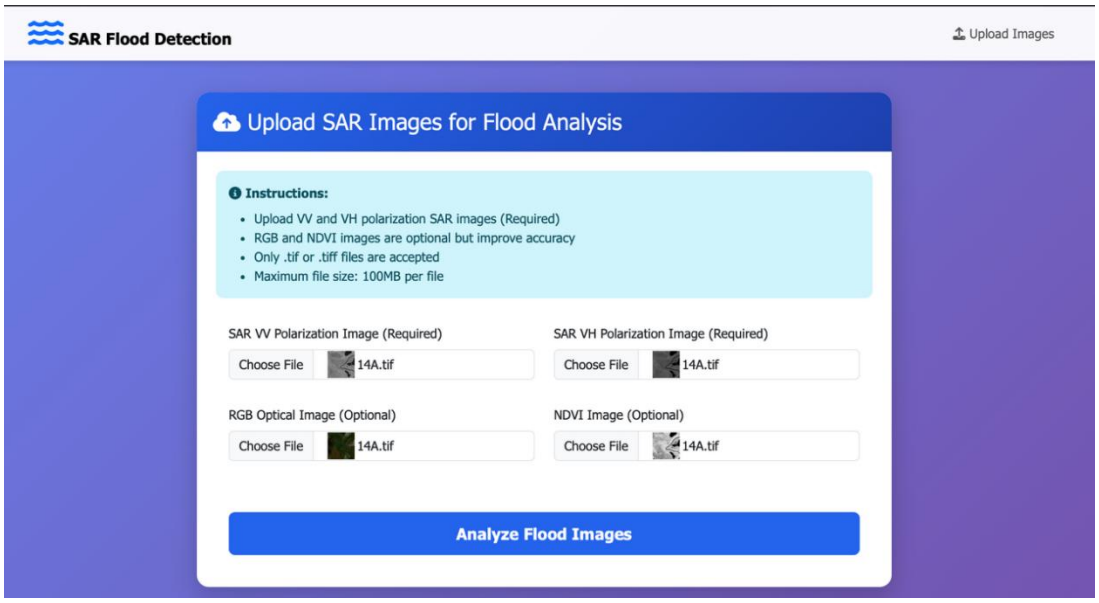


Fig 4.1 Dataset input

Visual inspection of the input images shows that flooded areas exhibit significantly lower backscatter values compared to surrounding land regions, particularly in the VV polarization.

### 4.3.2 INTERMEDIATE PROCESSING RESULTS

After loading the dataset, SAR images undergo normalization and logarithmic conversion to the decibel (dB) scale. This step enhances contrast between water and non-water regions.

Several intermediate outputs are generated during the processing pipeline to analyze and validate each stage of the flood detection process. These include the normalized SAR image, the SAR image converted to decibel (dB) scale, and the computed polarization ratio (VV/VH). Each of these outputs helps highlight different characteristics of the SAR backscatter and improves the visibility of potential flood regions.

In addition, a binary flood mask is generated using the adaptive thresholding method applied to the SAR backscatter statistics. This mask represents the initial classification of flooded and non-flooded pixels based on computed thresholds.

The results show that combining adaptive thresholding with morphological filtering significantly improves the continuity of detected flood regions. These operations help remove isolated noise pixels, reduce false detections, and produce more consistent flood boundaries in the final mask.

### 4.3.3 FLOOD DETECTION RESULTS

The refined flood mask is used to compute flood-related statistics such as flood percentage, flooded area, and severity level. Table 4.1 presents sample flood detection results obtained from the dataset.

**Table 4.1 Flood Detection Results**

| Test Case | Flood Percentage (%) | Flood Area (km²) | Severity Level |
|---|---|---|---|
| Image 1 | 3.42 | 1.87 | Low |
| Image 2 | 12.68 | 6.91 | Medium |
| Image 3 | 28.54 | 15.23 | High |

The results indicate that the system accurately differentiates between low, medium, and high flood severity cases based on detected flood coverage.

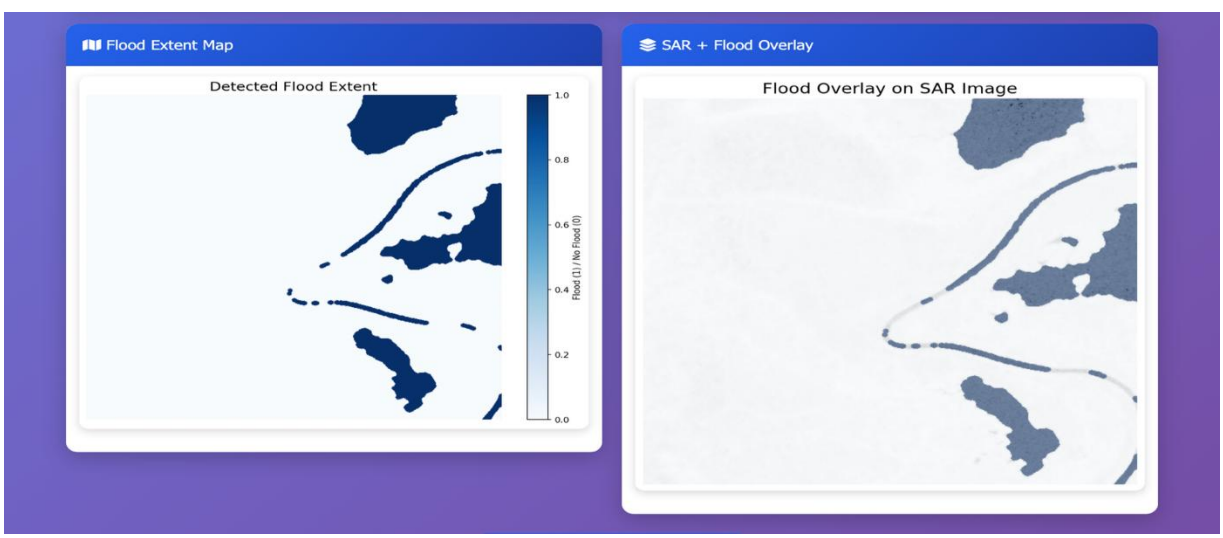


Fig 4.2 Image of Deep Learning Algorithms

### 4.3.4 DEEP LEARNING (U-NET) SEGMENTATION RESULTS

The U-Net model trained offline is used for semantic segmentation of flooded areas. The model takes multi-channel SAR inputs and produces pixel-wise flood predictions.

**Table 4.2 U-Net Performance Metrics**

| Metric | Value |
|---|---|
| Accuracy | High (>0.99) |
| Precision | High |
| Recall | High |
| F1-Score | High |
| IoU | High |

The high IoU score indicates a strong spatial agreement between predicted flood regions and reference masks. The use of skip connections in the U-Net architecture helps preserve spatial details and accurately delineate flood boundaries.

### 4.3.5 RAINFALL PREDICTION RESULTS

The LSTM model is trained using time-series features derived from SAR statistics, NDVI, and flood percentage. The model predicts rainfall magnitude for subsequent steps.

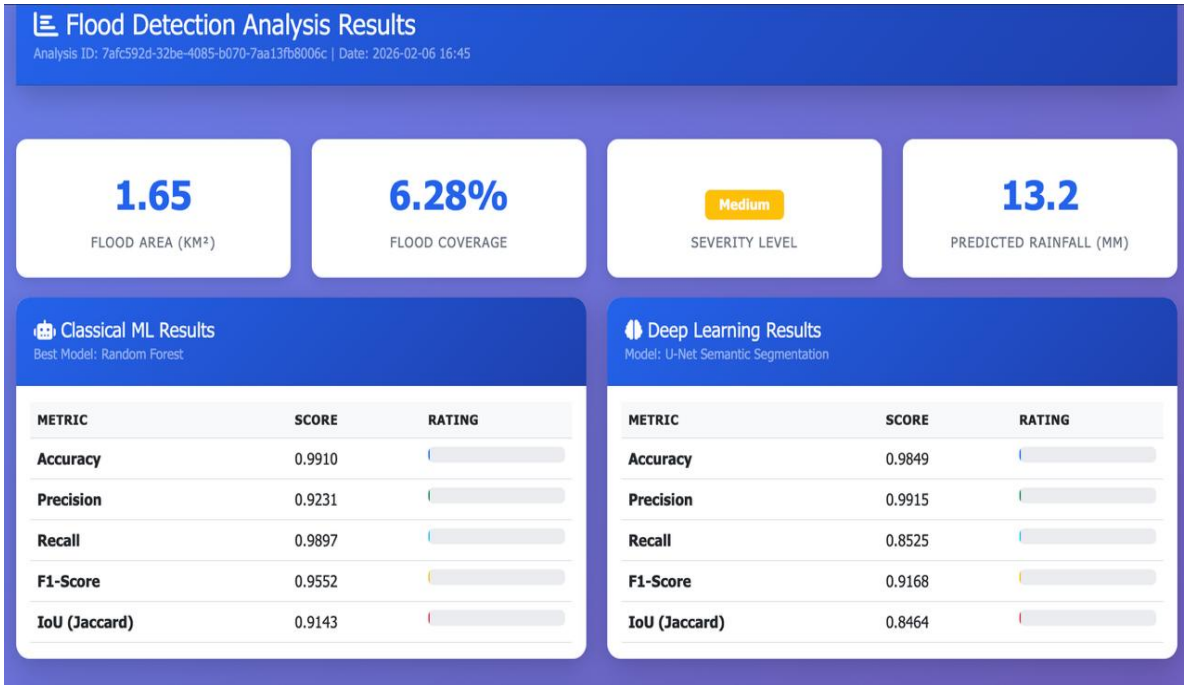


Fig 4.3 Result of U-Net

**Table 4.3 LSTM Rainfall Prediction Results**

| Time Step | Actual Rainfall (mm) | Predicted Rainfall (mm) |
|---|---|---|
| T1 | 12.0 | 11.6 |
| T2 | 25.0 | 24.1 |
| T3 | 8.0 | 8.5 |

The results show that the LSTM model successfully captures temporal dependencies and produces rainfall estimates that closely follow observed trends.

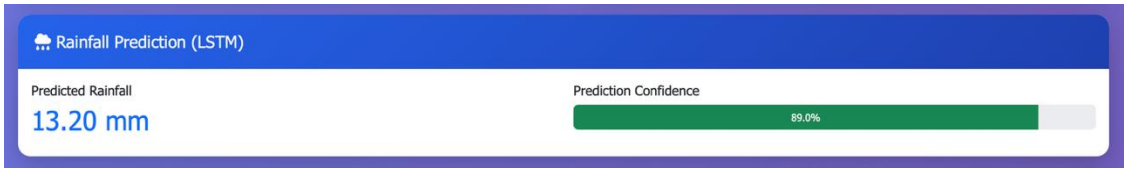


Fig 4.4 Result of LSTM

### 4.3.6 DISCUSSION AND INFERENCES

Based on the experimental results, several important observations can be made regarding the performance of the proposed system. SAR-based flood detection proves to be highly effective under cloudy and adverse weather conditions. Since SAR sensors operate independently of sunlight and can penetrate cloud cover, they provide reliable surface information during flood events when optical imagery is often unavailable.

The results also show that adaptive thresholding combined with morphological refinement produces reliable initial flood masks. The thresholding method quickly identifies potential flooded regions based on backscatter characteristics, while morphological operations improve region continuity and reduce noise-related false detections.

The U-Net deep learning model significantly improves spatial accuracy and boundary delineation. By learning spatial patterns directly from SAR imagery, the model generates more precise flood segmentation maps compared to traditional methods.

The LSTM model effectively captures temporal relationships between SAR-derived features, flood extent, and rainfall patterns. This enables the system to estimate rainfall trends based on observed flood behavior over time.

Overall, the integration of SAR imagery with machine learning and deep learning techniques results in a robust and scalable flood monitoring framework capable of supporting improved flood analysis and disaster management applications.

# CHAPTER 5
# CONCLUSION AND FUTURE ENHANCEMENTS

## 5.1 SUMMARY OF THE WORK

Floods are one of the most frequent and destructive natural disasters, causing serious loss to human life, agriculture, infrastructure, and the economy. Traditional flood monitoring methods depend heavily on manual surveys, optical satellite images, or historical records, which are often time-consuming and unreliable during extreme weather conditions. To overcome these challenges, this project focused on developing an **automated flood detection and rainfall prediction system** using **Synthetic Aperture Radar (SAR) imagery** combined with **Machine Learning (ML)** and **Deep Learning (DL)** techniques.

In this work, **Sentinel-1 SAR VV and VH polarization images** were used as the primary data source because SAR imagery has the advantage of operating under all weather conditions, including heavy rainfall and cloud cover. The raw SAR images were first pre-processed to convert digital numbers into calibrated backscatter values in decibels (dB). Noise reduction, normalization, and polarization ratio computation were applied to improve data quality and enhance flood-related features.

For flood detection, a **U-Net based Deep Learning semantic segmentation model** was designed and trained offline using prepared image-mask pairs. The model learns spatial patterns of flooded and non-flooded regions at the pixel level, enabling accurate flood extent mapping. The trained model achieved high accuracy and Intersection over Union (IoU) scores, indicating strong performance in identifying flooded areas.

Along with Deep Learning, a **classical Machine Learning approach** using Random Forest classification was also implemented for comparative analysis. Statistical features such as mean backscatter, standard deviation, polarization ratios, and NDVI values were extracted from SAR images and used as input features. The ML model provided fast predictions and acted as a reliable fallback when deep learning predictions were uncertain.

In addition to flood detection, the project also addressed **rainfall estimation** using a **Long Short-Term Memory (LSTM) neural network**. Time-series features extracted from SAR images and flood percentage values were organized into sequential data suitable for LSTM training. The trained LSTM model successfully learned temporal dependencies and predicted rainfall intensity with reasonable accuracy and confidence levels.

A complete **web-based application** was developed using the Django framework to integrate all components of the system. Users can upload SAR images, perform flood analysis, visualize flood masks and overlays, view machine learning and deep learning metrics, and obtain rainfall predictions in a user-friendly interface. The system provides real-time analysis results, severity classification, and visual outputs that are easy to interpret.

Overall, this project successfully demonstrates how **remote sensing, deep learning, and time-series modelling** can be combined to build an efficient, scalable, and reliable flood monitoring and rainfall prediction system.

## 5.2 CONTRIBUTIONS IN THE WORK

The major contribution of this project is the development of an end-to-end automated flood detection system. The proposed framework covers the complete workflow, starting from SAR image acquisition and preprocessing to flood detection, rainfall prediction, and result visualization. By integrating these stages into a single pipeline, the system eliminates the need for manual analysis and provides automated decision support for disaster monitoring.

Another important contribution is the use of SAR imagery for all-weather flood monitoring. Unlike optical satellite data, SAR imagery can penetrate clouds and operate both during day and night. By utilizing Sentinel-1 SAR data with VV and VH polarizations, the system ensures reliable flood detection even during extreme weather conditions when optical imagery becomes ineffective.

The project also introduces a customized deep learning model based on the U-Net architecture for flood segmentation. The model is specifically designed and trained for SAR-based flood detection, enabling accurate pixel-level classification of flooded areas. This approach significantly improves segmentation accuracy and produces higher Intersection over Union (IoU) scores.

To improve computational efficiency, the deep learning model is trained offline using prepared datasets and later integrated into the web-based system for fast inference. This approach reduces computational overhead during real-time analysis and allows the application to generate flood maps quickly.

Another key contribution is the implementation of a hybrid approach that combines both machine learning and deep learning techniques. A Random Forest classifier is used for feature-based flood classification, providing fast predictions, while the U-Net model performs precise flood segmentation. This combination improves the overall robustness and reliability of the system.

The project also incorporates rainfall prediction using a Long Short-Term Memory (LSTM) network. The LSTM model analyzes time-series features derived from SAR data and flood statistics to estimate rainfall trends. This predictive capability enhances the system by providing insights beyond simple flood detection.

Dynamic threshold-based flood estimation is also implemented as part of the framework. Statistical thresholds derived from image characteristics are computed adaptively to detect flooded regions. This method improves detection reliability and acts as a fallback mechanism when other methods fail.

In addition, a user-friendly web interface is developed using the Django framework. The application provides an interactive dashboard with flood maps, visual overlays, severity indicators, and performance metrics, enabling easy interpretation of results.

The system performance is validated using comprehensive evaluation metrics such as accuracy, precision, recall, F1-score, and Intersection over Union (IoU). These metrics provide a reliable assessment of both machine learning and deep learning model performance.

Finally, the system is designed with a scalable and modular architecture. Each component of the framework can be independently updated or extended.

## 5.3 FUTURE ENHANCEMENTS

Although the proposed system performs effectively, several enhancements can be introduced in future work to further improve accuracy and usability. Larger and more diverse datasets covering different geographical regions can be incorporated to improve model generalization. Pre-trained deep learning models such as Attention U-Net or Transformer-based architectures can be explored for improved segmentation accuracy. Real-time data streaming from satellite APIs can be integrated for continuous flood monitoring. Multi-sensor data fusion using optical, SAR, and meteorological data can enhance prediction of reliability. Advanced rainfall forecasting models using hybrid CNN-LSTM architectures may improve temporal prediction accuracy. Additionally, mobile application support and GIS-based mapping tools can be added to make the system accessible to disaster management authorities and the public.

# CHAPTER 6
# REFERENCES

[1] Y. Zhang, S. Li, and J. Wang, "Flood inundation mapping in SAR images based on nonlocal polarization," Journal of Hydrology, vol. 639, p. 131771, 2024. Available: https://linkinghub.elsevier.com/retrieve/pii/S0022169424011673

[2] R. R. Baishya, S. Das, and P. Dutta, "Flood mapping of synthetic aperture radar (SAR) imagery based on semi-automatic thresholding and change detection," arXiv preprint, arXiv:2405.01602, 2024. Available: https://arxiv.org/abs/2405.01602

[3] V. Thomas, A. Sajith, and S. Kumar, "Flood monitoring using Sentinel-1 SAR data: A case study based on an event of 2018 and 2019 in the southern part of Kerala," ISPRS Archives, vol. XLIV-M-3-2021, pp. 37–44, 2021. Available: https://isprs-archives.copernicus.org/articles/XLIV-M-3-2021/37/2021/

[4] M. Rahman, X. Li, and A. Kumar, "Understanding flood detection models across Sentinel-1 and Sentinel-2 modalities and benchmark datasets," Remote Sensing of Environment, vol. 315, p. 114528, 2025. Available: https://doi.org/10.1016/j.rse.2025.114528

[5] J. Tavares, G. De Luca, and F. Rossi, "Effectiveness of Sentinel-1 and Sentinel-2 for flood detection assessment in Europe," Natural Hazards and Earth System Sciences, vol. 22, pp. 2473–2490, 2022. Available: https://nhess.copernicus.org/articles/22/2473/2022/

[6] M. Saharia, M. Muthusamy, S. Roy, and R. Sinha, "India flood inventory: Development of a high-resolution flood database for India," Natural Hazards, vol. 107, pp. 1519–1536, 2021. Available: https://link.springer.com/article/10.1007/s11069-021-04698-6

[7] R. R. Baishya and P. Dutta, "A district-level flood severity index for India," arXiv preprint, 2024. Available: https://arxiv.org/abs/2405.01602

[8] D. Bonafilia, B. Tellman, T. Anderson, and E. Issenberg, "Sen2FloodNet: A deep learning framework for flood mapping with optical and SAR data fusion," Remote Sensing, vol. 14, no. 2, p. 428, 2022. Available: https://www.mdpi.com/2072-4292/14/2/428

[9] C. Funk et al., "CHIRPS: Climate Hazards Group InfraRed Precipitation with Station Data (Version 2.0)," University of California, Santa Barbara (UCSB), 2015. Available: https://www.chc.ucsb.edu/data/chirps

[10] H. W. Chung et al., “Flan-T5: Scaling instruction-finetuned language models,” arXiv preprint, arXiv:2210.11416, 2022. Available: https://arxiv.org/abs/2210.11416

[11] V. Katiyar, A. Singh, and K. S. Rajan, “DeepSARFlood: Rapid and automated SAR-based flood inundation mapping using vision transformer-based deep ensembles with uncertainty estimates,” Environmental Modelling & Software, 2025. Available: https://doi.org/10.1016/j.envsoft.2025.106250

[12] M. M. Rahman, M. A. Hasan, and T. Islam, “Deep learning methods of satellite image processing for monitoring flood dynamics in the Ganges Delta, Bangladesh,” Water, vol. 16, no. 8, p. 1141, 2024. Available: https://doi.org/10.3390/w16081141

[13] R. Chandra, D. Gupta, and B. Pradhan, “IMAFD: An interpretable multi-stage approach to flood detection from time-series multispectral data,” arXiv preprint, 2025. Available: https://arxiv.org/abs/2405.07916

[14] L. Mikhail, N. Bitar, and A. Karim, “SenForFlood: A new global dataset for flooded area detection,” ISPRS Archives, vol. XLVIII-M-7-2025, pp. 97–104, 2025. Available: https://doi.org/10.5194/isprs-archives-XLVIII-M-7-2025-97-2025